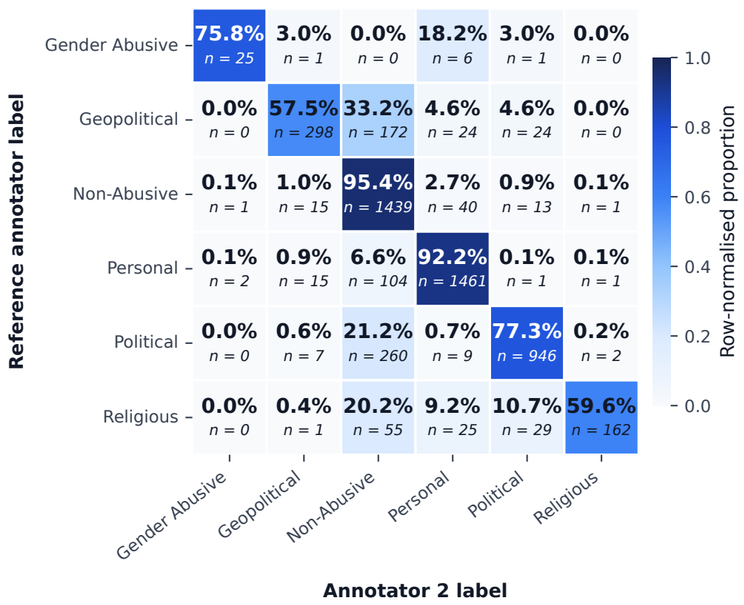
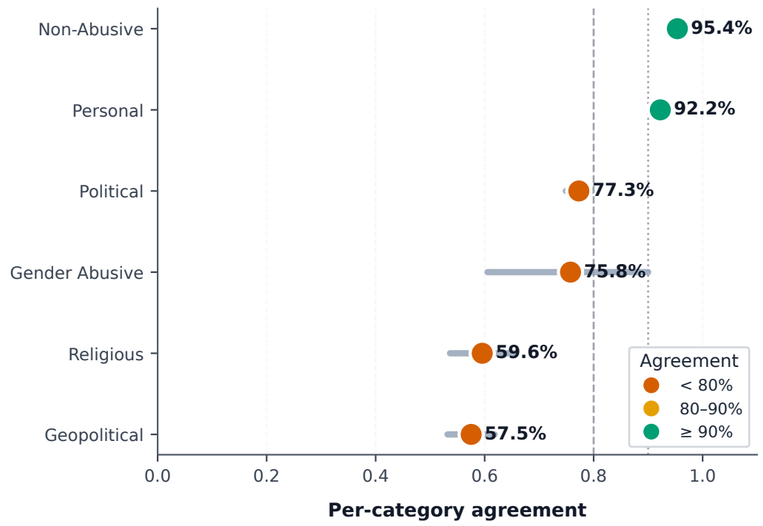
Inter-Annotator Reliability Assessment

Multiclass Cohen's κ = 0.786 • Substantial agreement • Observed agreement = 84.3% • N = 5,140 • 6 categories

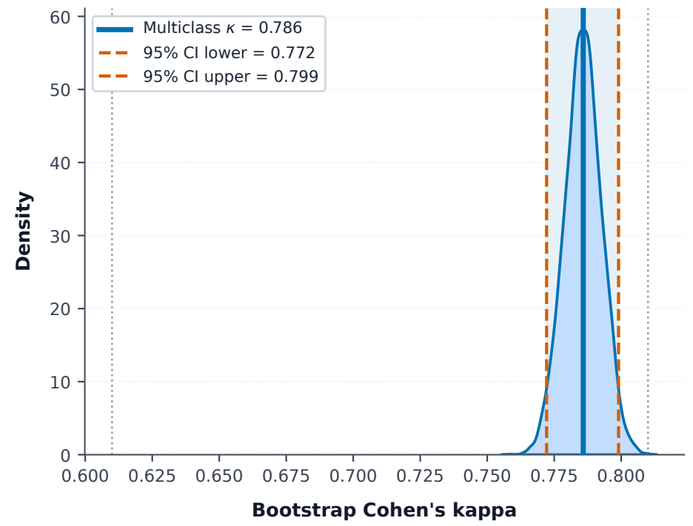
a Category-level agreement structure



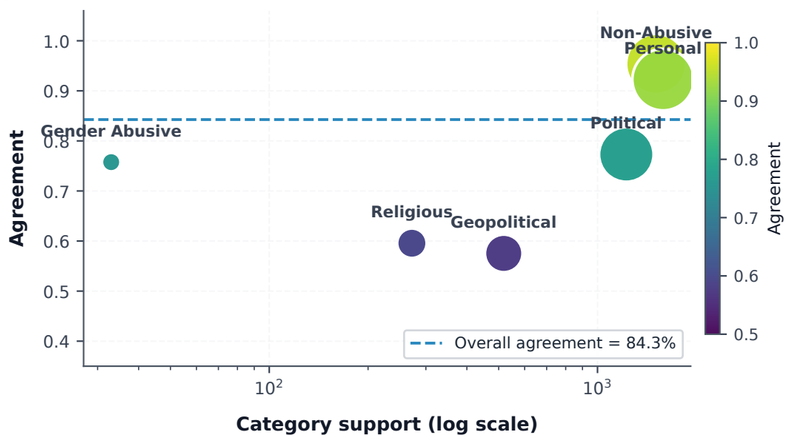
b Category-wise reliability with 95% CI



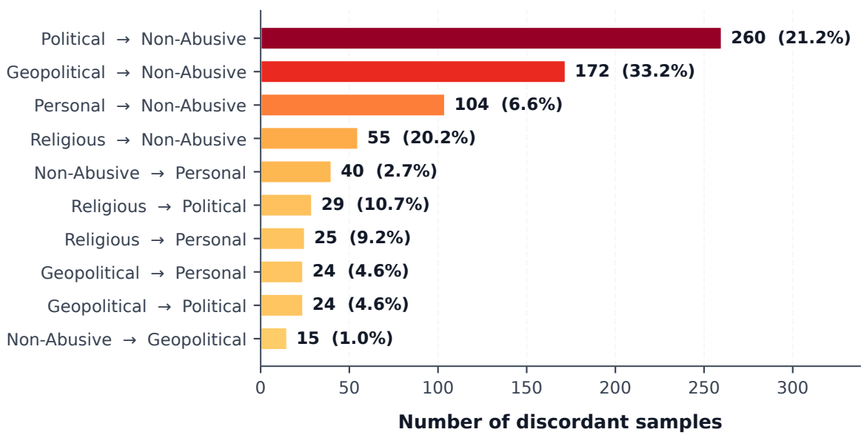
c Uncertainty in overall agreement 10,000 bootstrap resamples



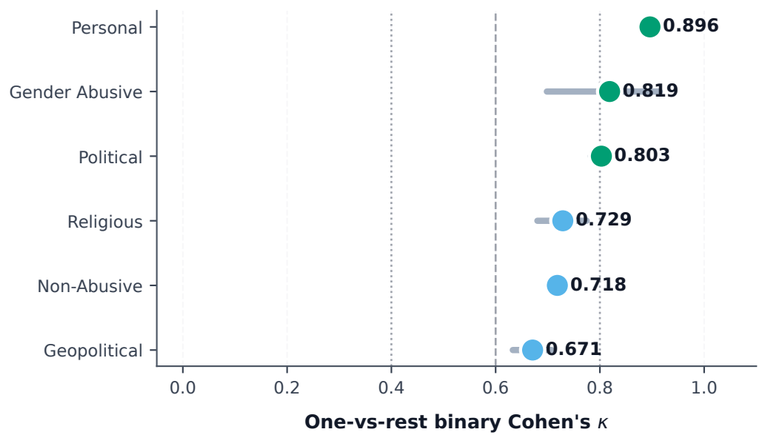
d Does category frequency affect agreement?



e Most frequent disagreement patterns 809 total discordant annotations



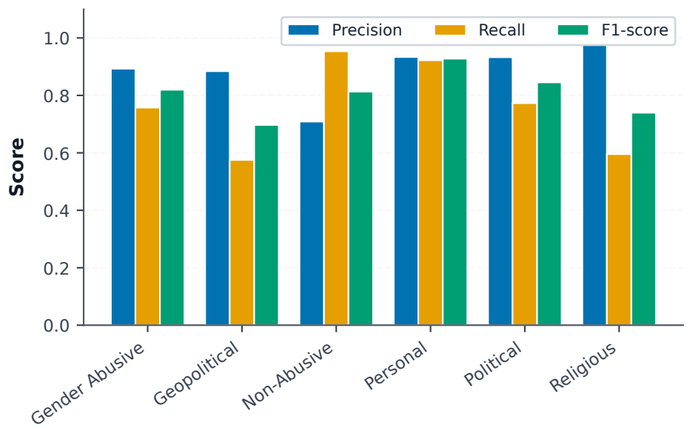
f Binary category-specific reliability with bootstrap 95% CI



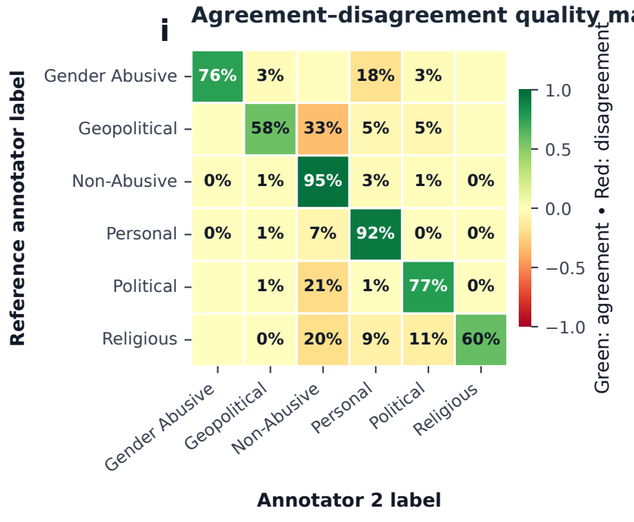
g Reliability scorecard

Samples	**5,140**
Categories	**6**
Observed agreement	**84.26%**
Balanced accuracy	**76.29%**
Macro F1-score	**0.807**
Multiclass Cohen's κ	**0.786**
Multiclass 95% CI	**[0.772, 0.799]**
Interpretation	**Substantial**

h Per-category precision, recall and F1



i Agreement–disagreement quality map



j Label distribution by annotator

k Observed agreement versus binary reliability

l Kappa interpretation framework



Multiclass κ = 0.786

95% CI: [0.772, 0.799]

Substantial agreement

Binary reliability was computed using a one-vs-rest decomposition, where each category was treated as the positive class and all remaining categories as the negative class. Bootstrap uncertainty was estimated using 10,000 resamples.